

Exploring Information Theory: Entropy, KL Divergence, Properties, Computations, and Applications in Statistics

Luai Al Labadi, Zhirui Chu, Nevena Ciganovic, Hanshuo Ma, Lucas Albert Noritomi-Hartwig, Jingwen Shi, Ying Xu

Department of Mathematical and Computational Science, University of Toronto Mississauga

Abstract

This poster aims to provide a comprehensive overview of key concepts in information theory, focusing on entropy, Kullback-Leibler (KL) divergence, their properties, computational methods, and applications in statistics.

Introduction to Information Theory

Information theory originally studies the transmission, processing, extraction, and utilization of information. In the case of communication of information over a noisy channel, the information is thought of as possible messages. The goal is to send, receive, and reconstruct these messages with a low probability of error, in spite of the channel noise. Information theory now has many relationships with other fields, including economics, computer science, communication theory, statistics, and mathematics. A key measure in information theory is entropy (Cover and Thomas, 1991).

Entropy

Entropy is a notion taken from thermodynamics, where it describes the uncertainty in the movement of gas particles. In statistics, entropy is considered as a measure of the uncertainty of a random variable.

Shannon Entropy

$$H(X) = H(f) = -\mathbb{E}[\ln(f(X))] = \begin{cases} -\sum_{x \in X} f(x) \ln(f(x)), & \text{if } f(x) \text{ is discrete} \\ -\int f(x) \ln(f(x)) dx, & \text{if } f(x) \text{ is continuous} \end{cases}$$

Joint and Conditional Entropy

For two discrete random variables, X and Y , with a joint distribution, $f(x, y)$:

The *joint entropy*, $H(X, Y)$, is given by

$$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} f(x, y) \ln(f(x, y)).$$

The *conditional entropy* of Y given X is

$$H(Y|X) = -\sum_{x \in X, y \in Y} f(x, y) \ln\left(\frac{f(x, y)}{f(x)}\right).$$

Main Properties:

$H(Y|X) = H(Y)$ if and only if X and Y are independent random variables.

Chain Rule: $H(X, Y) = H(X) + H(Y|X)$

Common Entropies

Distribution	PDF: $f(x)$	Entropy: $H(x)$
Normal Distribution	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\ln(\sigma\sqrt{2\pi}e)$
Exponential	$\xi e^{-\xi x}$	$1 - \ln \xi$
Uniform	$\frac{1}{b-a}$	$\ln(b-a)$

KL Divergence and Cross Entropy

KL divergence is a type of statistical distance, a measure of how one probability distribution, f , is different from a second reference probability distribution, g .

$$D_{KL}(f||g) = \begin{cases} \sum_{x \in \mathcal{X}} f(x) \ln\left(\frac{f(x)}{g(x)}\right), & \text{if } f(x) \text{ and } g(x) \text{ are discrete} \\ \int f(x) \ln\left(\frac{f(x)}{g(x)}\right) dx, & \text{if } f(x) \text{ and } g(x) \text{ are continuous} \end{cases}$$

It is known that $D_{KL}(f||g) \geq 0$, where $D_{KL}(f||g) = 0$ if and only if $f = g$ almost surely. This property makes it a useful tool in many areas of machine learning and information theory, such as hypothesis testing, model selection, and clustering. The cross-entropy is an information measure that can be regarded as an error metric between two distributions:

$$S(f, g) = D_{KL}(f||g) + H(f)$$

Measures of Sample Entropy

In most practical scenarios, the true PDF f is unknown, and hence, we need to estimate the entropy from the available data, which can be a challenging task. Vasicek (1976) derived the expression for $H(X)$ in terms of the inverse of the distribution function F , given by

$$H(X) = \int_0^1 \log\left(\frac{d}{dt}F^{-1}(t)\right) dt.$$

If $x_1, \dots, x_n$ is a sample from F , then an estimator of $H(X)$ is given by Vasicek's estimator, introduced in 1976 and later studied by Ebrahimi et al. in 1994:

$$H(m, n) = \frac{1}{n} \sum_{i=1}^n \ln \frac{x_{i+m} - x_{i-m}}{c_i m/n},$$

where

$$c_i = \begin{cases} 1 + \frac{i-1}{m}, & 1 \leq i \leq m, \\ 2, & m+1 \leq i \leq n-m, \\ 1 + \frac{n-i}{m}, & n-m+1 \leq i \leq n, \end{cases}$$

where m is a positive integer smaller than $n/2$, and $X_{(1)} \leq \dots \leq x_{(n)}$ are the order statistics of $x_1, \dots, x_n$ with $x_{(i-m)} = x_{(1)}$ if $i \leq m$ and $x_{(i+m)} = x_{(n)}$ if $i \geq n-m$.

How to choose m ? $m = \lfloor \sqrt{n} + 0.5 \rfloor$ (Al-Labadi et al., 2021).

To assess the efficiency of the entropy estimator, we analyzed 5000 samples of sizes $n = 20, 50, 100$ from both the standard normal and exponential distributions. The results, along with the corresponding mean squared errors, are presented in Tables 1 and 2.

Normal(0, 1)				
n	m	Exact	Est	MSE
20	4	1.4189	1.2555	0.0592
50	7	1.4189	1.3522	0.0161
100	10	1.4189	1.3931	0.0060

Table 1: Normal Distribution

Exp(1)				
n	m	Exact	Est	MSE
20	4	1.0000	0.9080	0.0699
50	7	1.0000	0.9743	0.0237
100	10	1.0000	0.9869	0.0112

Table 2: Exponential Distribution

Application

One major application of the KL divergence is model checking. Assume we have $x_1, \dots, x_n \sim F$ where F is an unknown distribution. We want to test the null hypothesis $H_0 : F = F_0$, where F_0 is a known distribution with PDF f_0 . It can be shown that $D_{KL}(f||f_0)$ may be estimated by

$$d = \widehat{D}_{KL}(f||f_0) = \hat{H}(X) - \hat{S}(f, f_0) = H(m, n) - \frac{1}{n} \sum_{i=1}^n \log f_0(x_{(i)}).$$

To calibrate the result, McCulloch proposed the following calibration function: $q(k) = (1 + (1 - e^{-2f})^{1/2})/2$. As q approaches 0.5, the two distribution are more likely to be the same. As an example, we generated a sample of size $n = 50$ from the standard normal distribution and considered different cases of F_0 . The results are reported in Table 3.

F_0	$\hat{d}$	q
$N(0, 1)$	0.0464	0.6488
$N(3, 1)$	4.3019	1.0000
t_1	0.3222	0.8446

Table 3: Model Checking

References

- [1] Al-Labadi, L., Patel, V., Vakiloroyaei, K., and Wan, C. (2021b). Kullback-Leibler divergence for Bayesian nonparametric model checking. *Journal of the Korean Statistical Society*. <https://doi.org/10.1007/s42952-020-00072-7>
- [2] Cover, T. M., and Thomas, J. A. (1991). *Elements of Information Theory*, 2nd edition. New York: John Wiley & Sons, Inc.
- [3] Ebrahimi, N., Pflughoeft, K., and Soofi, E. S. (1994). Two measures of sample entropy. *Statistics & Probability Letters*, 20, 225-234.
- [4] McCulloch, R. (1989). Local prior influence. *Journal of the American Statistical Association*, 84, 473-478.
- [5] Vasicek, O. (1976). A test for normality based on sample entropy. *Journal of Royal Statistical Society B*, 38, 54-59.

Contact information: lucas.noritomi@utoronto.ca